

websites

Documentation: Websites and Online Resources

This document outlines the key online resources used for data acquisition, software distribution, and visualization for the lunar elemental mapping project.

1. ISRO Planetary Science Data Archive (PRADAN)

- **Website:** ISRO - Planetary Data Archival and Delivery
 - **URL:** <https://pradan.issdc.gov.in/>
 - **Purpose in this Project: Primary Source for All Scientific Data.**
 - **Specific Resources Used:**
 - **Chandrayaan-2 CLASS Level-1 Data:** These are the raw, calibrated `.fits` files containing the lunar X-ray spectra. Each file represents a single, short observation of a specific footprint on the Moon.
 - **Chandrayaan-2 XSM Level-1 Data:** These are the daily `.fits` files containing the per-second solar X-ray spectra.
 - **Official Calibration Files:** This archive is also the source for all the necessary calibration files for both CLASS and XSM, including the RMF/RSP (Response), ARF (Ancillary), and BKG (Background) files.
 - **Role in the Pipeline:** The PRADAN archive provided the **fundamental input for the entire project**. The data downloaded from this site was processed in **Phase 1** (Data Preparation) and was the subject of the scientific analysis throughout **Phase 2** (The Batch Run).
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2. LROC Science Operations Center (NASA / ASU)

- **Website:** Lunar Reconnaissance Orbiter Camera (LROC) Science Operations Center, hosted by Arizona State University.
- **URL:** <https://lroc.sese.asu.edu/images/downloads>
- **Purpose in this Project: Source for the High-Resolution Lunar Base Map.**

- **Specific Resources Used:**
 - **LRO WAC Global Morphology Mosaic:** This is the official global map of the Moon's surface brightness (albedo) created from images taken by the Wide Angle Camera (WAC) aboard NASA's Lunar Reconnaissance Orbiter.
 - We specifically used a PNG version of this map (e.g., `WAC_GLOBAL_BD100_2048.PNG` or the custom-generated 16 pixels-per-degree version).
 - **Role in the Pipeline:** This map served as the visual background for the final scientific plots created in **Phase 3**. The `plot_abundance_map.py` script overlaid the calculated elemental abundance data on top of this albedo map to provide geographic context.
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3. HEASARC (NASA)

- **Website:** High Energy Astrophysics Science Archive Research Center (HEASARC)
 - **URL:** <https://heasarc.gsfc.nasa.gov/>
 - **Purpose in this Project: Source for Essential Scientific Analysis Software.**
 - **Specific Resources Used:**
 - **HEASOFT:** This is a large suite of software for high-energy astrophysics. We used it to get its primary component, `XSPEC`.
 - **XSPEC:** The global standard for X-ray spectral fitting.
 - `pyxspec`: The official Python interface for `XSPEC`, which is included in the HEASOFT installation.
 - **Role in the Pipeline:** The `XSPEC` engine, controlled via the `pyxspec` library in `run_analysis_batch.py`, was the **core of the scientific modeling in Step 2.1.2**. It performed the critical **forward-fitting** required to calibrate the raw XSM solar spectrum, a step that is impossible to do correctly without this specialized software.
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4. The Python Package Index (PyPI)

- **Website:** The Python Package Index (PyPI)
- **URL:** <https://pypi.org/>
- **Purpose in this Project: Source for All Open-Source Python Libraries.**
- **Specific Resources Used:**
 - `pandas` , `numpy` , `astropy` , `scipy` , `matplotlib` , `plotly` , `healpy` .
- **Role in the Pipeline:** These libraries are the **fundamental building blocks of the entire custom analysis pipeline**. They were installed using the `pip` command within the project's virtual environment. Their roles ranged from data handling and file I/O (`pandas` , `astropy`) to numerical computation and fitting (`numpy` , `scipy`) and all final scientific visualizations (`matplotlib` , `plotly` , `healpy`).